Sahaj Bhadja

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Summary

Backend Software Engineer pursuing an MSCS at USC with 2+ years building **Python/Node.js APIs**, **third-party integrations**, and **AWS-backed services**, with experience in database design, data-flow automation, and AI-assisted workflows for customer-facing platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built **Python backend services and FastAPI endpoints** for 500+ users with multi-tenant data design, optimizing query patterns and indexing to cut request latency from 1.2s to 350ms for reliable performance.
- Engineered **ETL integrations and data-flow pipelines** across FastAPI services, databases, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, and CI/CD releases at 99%+ reliability.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built and maintained **backend APIs and integrations** (Flask, REST) handling 10K+ daily requests, connecting third-party LLM providers behind a switchable architecture with cost-tracking and automated CI/CD releases.
- Leveraged **AI tools and multimodal models** to automate workflows, reducing turnaround **60%** while collaborating with front-end developers and stakeholders to ship reliable production features for 150+ users.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built and optimized **server-side APIs and microservices** with AWS Glue and S3, improving response times **30%** through pagination, field filtering, indexing, and caching for scalable third-party data integrations.
- Maintained **CI/CD pipelines and 20+ test suites** (Jenkins, PyTest), documented systems and integrations, and increased pipeline reliability **68%** while collaborating with Microsoft/GitHub stakeholders in Agile sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed **REST APIs and microservices** (Node.js, Express) with WebSocket sync across eight workflows for 2M+ records, integrating Azure SQL and third-party services for reliable, automated data flow.
- Automated **Docker-based deployments** via Jenkins CI/CD on Azure App Services, cutting release time **50%** and improving stability, with Sentry and Azure Monitor for backend observability and troubleshooting.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built **Flask APIs and webhook integrations** for a cross-platform booking and payments platform across 50K+ records, connecting third-party payment gateways with real-time Firebase sync and sub-200ms responses.
- Automated **refund and payment workflows** through idempotent webhook handling, reducing manual handling **70%**, and mapped 12+ backend integration points while documenting architecture for client stakeholders.

Skills

Languages: Python, JavaScript/TypeScript, Java, SQL, Bash
Backend & APIs: FastAPI, Flask, Django, Node.js, Express.js, REST APIs, Webhooks, Microservices
Cloud & Infrastructure: AWS (S3, RDS, Glue, ECS, Lambda), Azure, Docker, Kubernetes, CI/CD, Terraform, Jenkins
Databases: PostgreSQL, MySQL, MongoDB, DynamoDB, Azure SQL, Database Design, Query Optimization
Frontend & Integrations: React.js, Next.js, JAMStack, CRM Integrations, Marketing Automation, Third-Party API Integrations, HTML/CSS
Practices & AI: AI/LLM Tools, Performance Optimization, Technical Documentation, Agile

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Built and deployed a **containerized REST API** on AWS (CodePipeline, ECS/Fargate) with automated CI/CD releases and validation checks, achieving 87% accuracy and production-grade reliability and scalability.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*
- Built an **AI-powered data workspace** across Java and TypeScript supporting 10+ databases, with schema-aware querying and Docker-backed deployment to accelerate backend data integration and management workflows.

PiXMod Live | *Next.js, MongoDB, Stripe, Transformers*
- Built an **AI-powered Next.js SaaS web platform** with Stripe billing and MongoDB-backed workflows, delivering a customer-facing experience that improved premium conversion **18%** through polished, scalable design.